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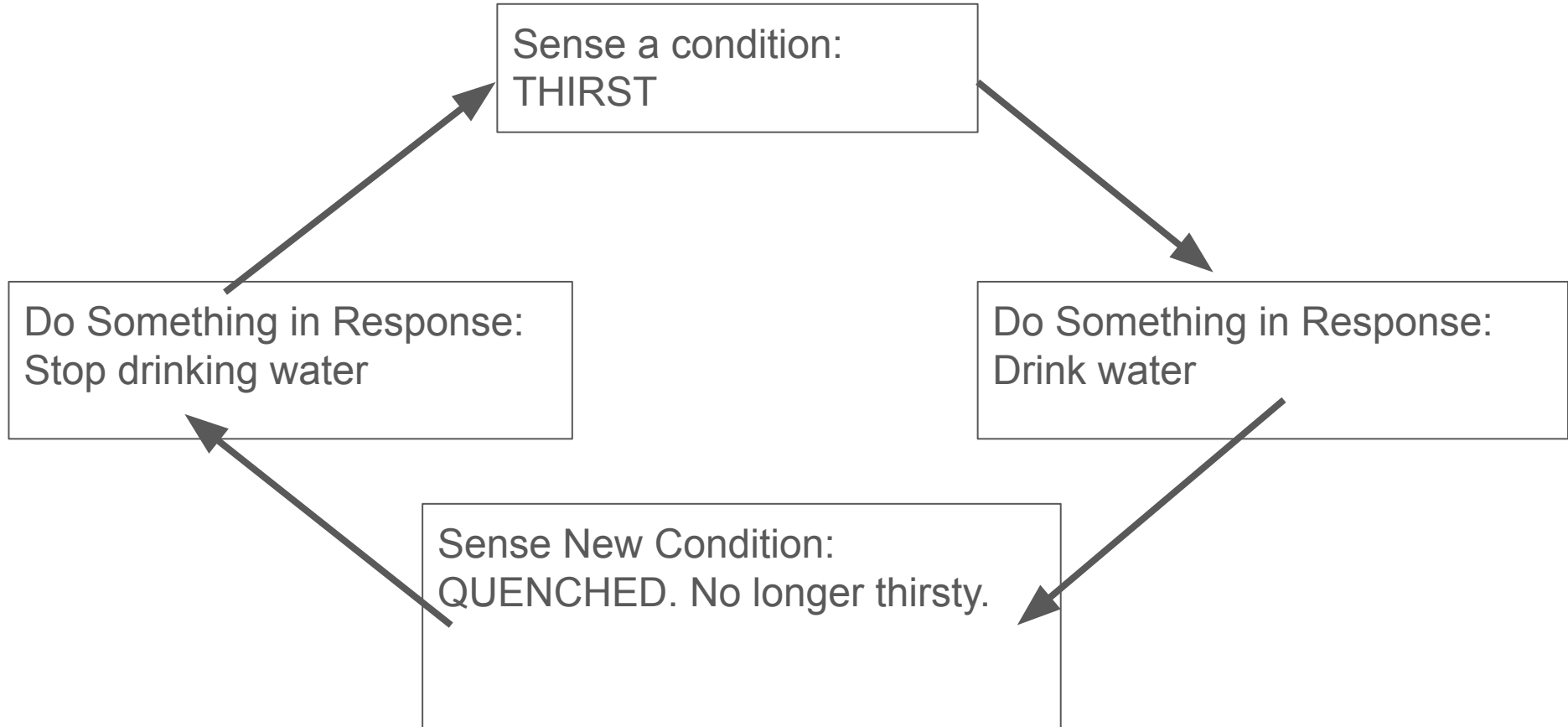
Homeostasis: Systems in the body work to maintain a balance of internal conditions so that all cells in the body have their needs met.

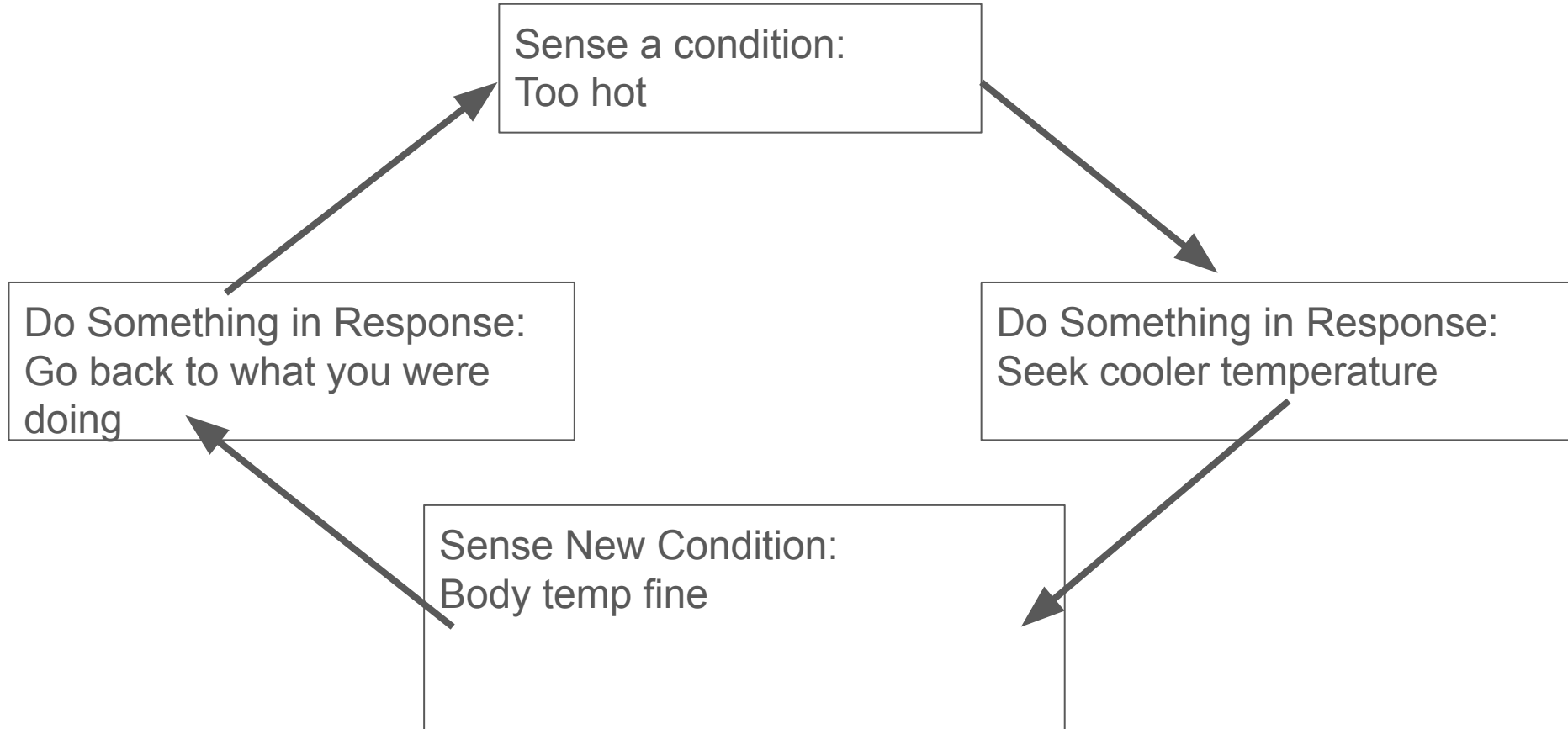
Internal conditions like:

- Temperature
- Water
- Food
- Nutrients
- Waste

Your body is constantly working to maintain homeostasis. Running, jumping, sitting in an air conditioned room, drinking a soda... it changes your internal conditions.

Activities that use a lot of energy
(such as fighting an illness)





Nutrients:

Carbohydrates: primary food molecule (simple like sugar, complex like fiber and 'whole grain')

Proteins: growth and repair, can also be used for energy

Fats: energy storage. growth, hormones, nerve stuff, also energy

Vitamins: used in various chemical reactions

Minerals: used in various chemical reactions

Water: important for everyday function of cells

Absorption

the body taking in
nutrients

Digestion

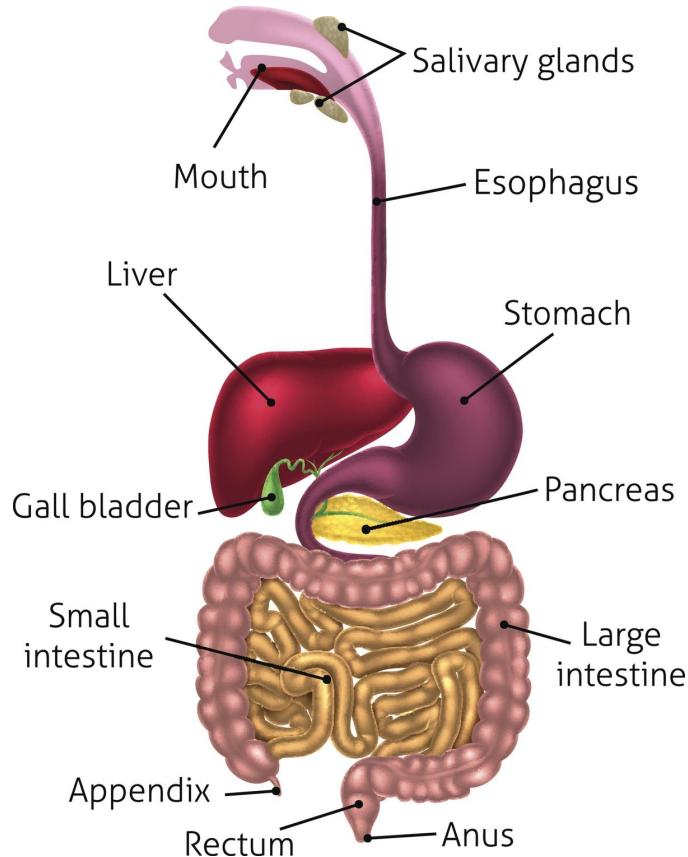
breaking down food
particles into absorbable
nutrients

Mechanical Digestion

Physically breaks food into small pieces, allows chemical digestion to happen faster

Chemical Digestion

break down food particles into nutrients that can actually be absorbed



Digestion

Mechanical
Stomach, esophagus,
mouth

Chemical
Pancreas, small
intestine, liver (+gall
bladder), salivary gland,
stomach

Absorption

Small intestine – villi increase surface area for
absorption

Large intestine – water and some vitamins

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Tri1 Review Questions

Topic Roadmaps

Learning Objectives

outline big picture concepts for each lesson

Vocabulary

bolded = core vocab, unbolded = additional terms

Questions

specific questions (ones that might show up on a test!)

How to use your Topic Roadmaps

Answer the questions in complete sentences, diagrams, or charts. Use as much academic vocabulary as possible.

Check that you can define or provide an example of each vocabulary term. Identify terms that are still hazy to you.

Complete or discuss the learning objectives.

November 4, 2025

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Tri1 Review Questions

Earthworm Dissection

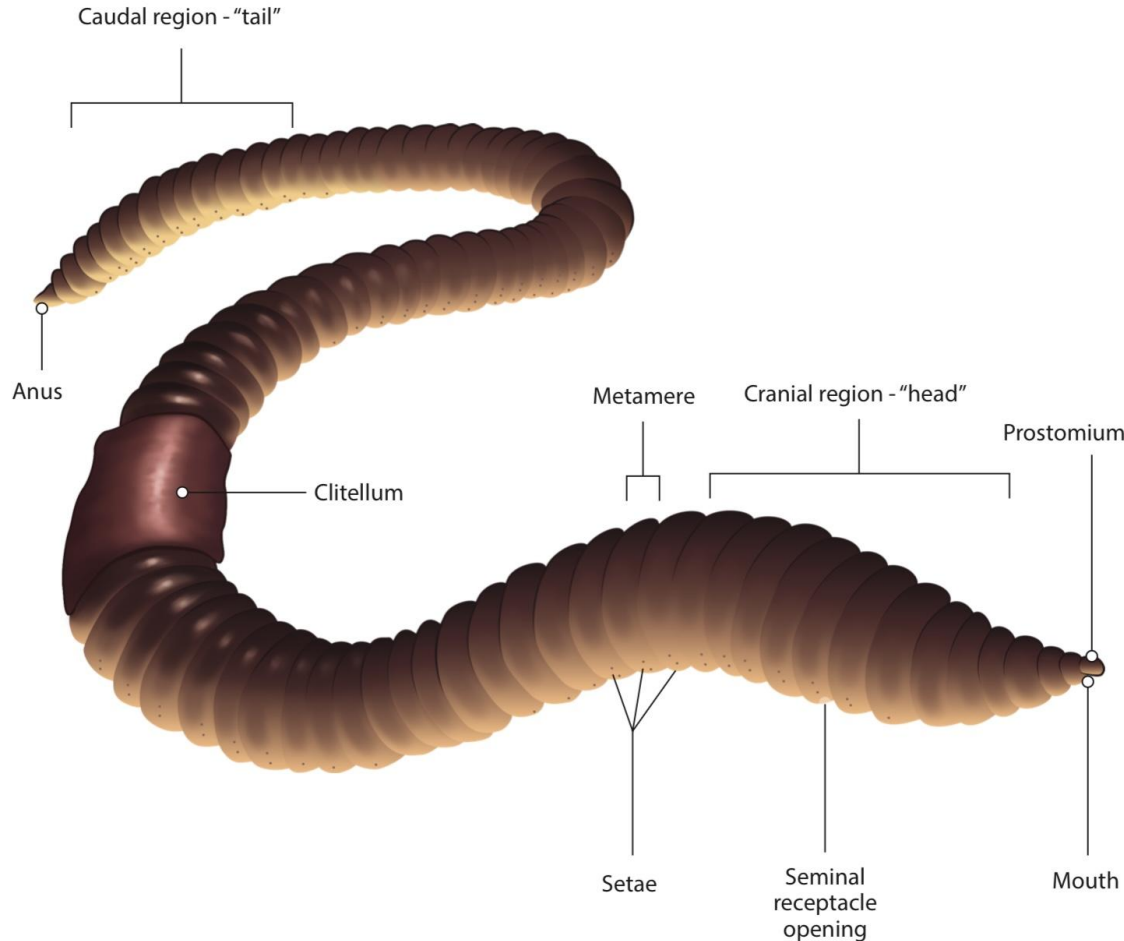
Lab Safety

- wear gloves when handling the specimen
- keep specimen on the dissection tray at all times
- only use dissection tools on the specimen
- do not make cuts before instructed to

Materials (per pair)

- styrofoam tray
- paper towel
- 2 probes
- 1 pair of scissors
- 4 pins
- gloves (2 per person)

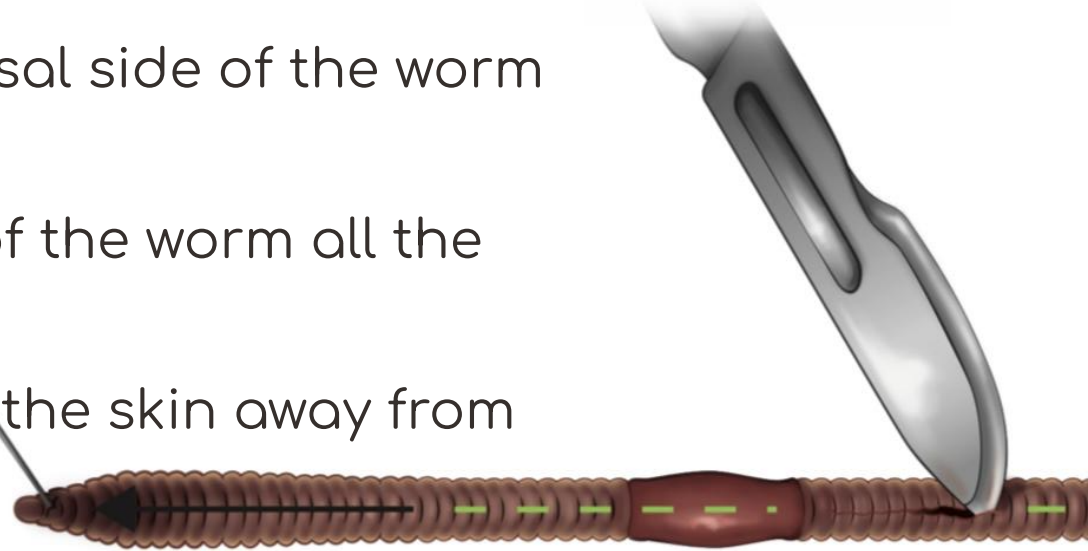
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**ORIENT YOUR
WORM**
Find the head
Dorsal (dark)
side up

Dissection Procedure:

1. Make a snip in the dorsal side of the worm below the clitellum.
2. Carefully cut the skin of the worm all the way up to the head.
3. Use the probes to pull the skin away from the internal organs.
4. Pin back the skin to reveal the organs.

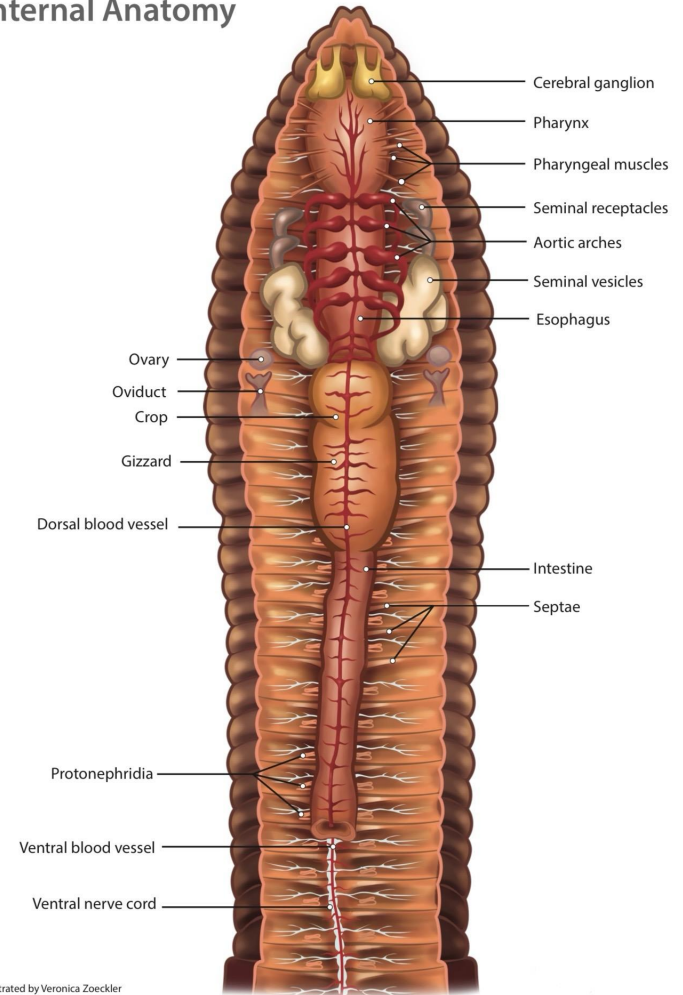


Earthworm Dissection

Discussion Questions:

1. How complex is the organization of the worm? Is it more or less complex than a human?
2. In what ways is the worm digestive system similar to humans? How is it different?

Internal Anatomy



Illustrated by Veronica Zoeckler

Double Loop Diagram

- Heart, lungs, body
- Label **oxygenated** and **deoxygenated** blood
- Arteries go away from the heart
- Veins return to the heart
- Capillaries are where gas exchange happen

